Two spins coupled to an XXZ bulk

Reduced channels and quantum-state texture

Spin transport project

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Two-probe reduced dynamics

Two spin-1/2 probes, A and B, attach to opposite ends of a finite XXZ bulk. The total Hamiltonian drives the dynamics; we follow only the probes.

- **Ising contacts:** the probes act as conserved controls on the bulk. The result is an exact, time-dependent dephasing (Schur) channel.
- **Exchange contacts:** spin flows between probes and bulk. Magnetization conservation splits the channel into five blocks.
- **Texture:** a single probe input, $|{+}{+}\rangle$, and a single measurement setting.

The slides give results and their meaning. All derivations are in the accompanying report.

Hamiltonian and normalization

N bulk spins, $N+2$ total. Physical order: $A, 1, \ldots, N, B$.

$$S^\alpha = \frac{\sigma^\alpha}{2}, \quad S^\pm = S^x \pm i S^y, \quad \hbar = 1.$$

$$H_E = J \sum_{j=1}^{N-1} \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z \right).$$

$$\begin{aligned} V_{\mathrm{ZZ}} &= g_A S_A^z S_1^z + g_B S_N^z S_B^z, \\ V_{\mathrm{ex}} &= \frac{g_A}{2}(S_A^+ S_1^- + S_A^- S_1^+) + \frac{g_B}{2}(S_N^+ S_B^- + S_N^- S_B^+), \\ H^{(r)} &= H_E + V_r, \qquad r \in \{\mathrm{ZZ}, \mathrm{ex}\}. \end{aligned}$$

J, g_A, g_B have energy units. These are alternative contact models, with no direct A–B interaction and no applied fields in the reference model.

A fixed product preparation defines a channel

Group the Hilbert space as $\mathcal{H}_{AB} \otimes \mathcal{H}_E$. Fix the same bulk density matrix ω_E for every probe input:

$$\rho(0) = \rho_{AB}(0) \otimes \omega_E, \qquad \boxed{\Lambda_t(X) = \mathrm{Tr}_E[U_t(X \otimes \omega_E)U_t^\dagger], \quad U_t = e^{-iH^{(r)}t}.}$$

- Λ_t is completely positive and trace preserving (CPTP) on the four-dimensional two-probe space.
- ω_E may be pure or mixed, thermal or not; changing it changes the channel.
- Exact unitary dynamics plus a partial trace: no weak-coupling or Markov approximation.
- Initially correlated probe–bulk states have reduced trajectories, but no universal channel.

Derivation: report, Sec. 2.

Ising coupling: the probes control the bulk

Both probe spins are conserved, $[H^{(\mathrm{ZZ})}, S_A^z] = [H^{(\mathrm{ZZ})}, S_B^z] = 0$. For each configuration $a = (s_A, s_B) \in \{00, 01, 10, 11\}$ the bulk feels a different boundary field:

$$H_a = H_E + g_A s_A S_1^z + g_B s_B S_N^z, \qquad \boxed{U_t = \sum_a |a\rangle\langle a| \otimes e^{-iH_a t}.}$$

- Four conditional bulk evolutions start from one common ω_E.
- The probes may occupy all four branches coherently.
- Exact for any bulk Hamiltonian on E, even with transverse fields.

Derivation: report, Sec. 3.1.

The reduced state: frozen populations, evolving coherences

For an arbitrary probe input $\rho_{AB}(0) = (r_{ab})$,

$$\boxed{[\rho_{AB}(t)]_{ab} = \Gamma_{ab}(t)\, r_{ab}, \qquad \Gamma_{ab}(t) = \mathrm{Tr}_E\left[\omega_E\, U_b(t)^\dagger U_a(t)\right].}$$

- $\Gamma_{aa} = 1$: the four populations never change.
- Γ_{ab} $(a \neq b)$ is the overlap of two conditional bulk histories. For a pure bulk, $|\Gamma_{ab}|^2$ is a conditional Loschmidt echo; the channel also needs its phase.
- Only six complex numbers evolve. They depend only on the boundary difference $H_a - H_b = g_A \Delta s_A\, S_1^z + g_B \Delta s_B\, S_N^z$.
- At $t = 0$, $\dot{\Gamma}_{ab}$ is purely imaginary: the motion starts coherently, and dephasing appears only later.

Derivation: report, Secs. 3.2 and 3.4.

A legitimate dephasing channel

$$\boxed{\Lambda_t(X) = \Gamma(t) \circ X} \qquad (\circ : \text{ entrywise product})$$

- Γ is a correlation matrix: $\Gamma \succeq 0$, unit diagonal, $|\Gamma_{ab}| \leq 1$. Hence CPTP and unital for every ω_E.
- Choi state $J_t = W\,(\Gamma/4)\,W^\dagger$, $W|a\rangle = |aa\rangle$: rank ≤ 4.
- Kraus operators are diagonal: $D_k = \sqrt{\lambda_k}\ \mathrm{diag}(v_k)$ from $\Gamma = \sum_k \lambda_k v_k v_k^\dagger$.
- Not depolarizing; not, in general, a product of single-probe channels.
- Dephasing includes coherent phases and revivals. It need not be irreversible.

Derivation: report, Secs. 3.3, 3.5 and 3.7.

The $|{+}{+}\rangle$ output is the whole channel

Since $|{+}{+}\rangle\langle{+}{+}|$ has all entries equal to $1/4$,

$$\boxed{\rho_f(t) = \Lambda_t(|{+}{+}\rangle\langle{+}{+}|) = \tfrac{1}{4}\Gamma(t), \qquad \Lambda_t(X) = [4\rho_f(t)] \circ X.}$$

- One input preparation, plus tomography of its six output coherences, reconstructs the channel.
- Populations are fixed at $1/4$, so eight local Pauli settings $(\alpha, \beta) \neq (z, z)$ suffice.
- Time derivatives of the same output give the time-local generator.
- Special to the Ising family: it fails for general two-probe channels.

Derivation: report, Sec. 3.6.

When can the channel be inverted?

$$\det \Lambda_t = \prod_{a<b} |\Gamma_{ab}(t)|^2.$$

- Invertible iff no coherence factor vanishes. Then the time-local generator is entrywise:

$$\dot{\rho}_{ab} = \eta_{ab}\rho_{ab}, \qquad \eta_{ab} = \frac{\dot{\Gamma}_{ab}}{\Gamma_{ab}} = -\kappa_{ab} + i\omega_{ab}.$$

- κ_{ab} is the observed fractional damping rate of one coherence; ω_{ab} is its phase rate.
- Intermediate map $\Lambda_t \Lambda_s^{-1}$: Schur multiplication by $\Gamma_{ab}(t)/\Gamma_{ab}(s)$. It is CPTP iff that matrix is $\succeq 0$.
- A coherence that vanishes and later revives rules out any intermediate map through that time.

Derivation: report, Secs. 3.8 and 4.1.

Canonical master equation: at most three rates

At every invertible time, the canonical form of Hall *et al.* [1] is

$$\dot{\rho} = -i[H_{\mathrm{eff}}, \rho] + \sum_{k=1}^{3} \gamma_k(t)\Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\}\Big).$$

- H_{eff} and all L_k are diagonal combinations of $\sigma_A^z,\ \sigma_B^z,\ \sigma_A^z\sigma_B^z$. No population transfer.
- Twelve of the fifteen canonical rates vanish identically.
- H_{eff} is a bulk-induced configuration energy, which can include an effective $\sigma_A^z\sigma_B^z$ coupling.
- Everything follows in closed form from the six $\eta_{ab}(t)$ alone.

Derivation: report, Secs. 4.2–4.8.

How the canonical data are obtained

Let $F_\mu = \frac{1}{2}(\sigma_A^z,\ \sigma_B^z,\ \sigma_A^z\sigma_B^z)$ and G be the 4×3 table of their eigenvalues $\pm\frac{1}{2}$ ($G^{\mathsf{T}} G = I_3$, $G^{\mathsf{T}}\mathbf{1} = 0$). Then

$$\boxed{\mathrm{d} = G^{\mathsf{T}} \eta\, G, \qquad h_a = -\tfrac{1}{4} \sum_b \operatorname{Im} \eta_{ab}, \qquad \gamma_k = \operatorname{spec} \mathrm{d}, \quad L_k = \textstyle\sum_\mu V_{\mu k} F_\mu.}$$

- Only a 3×3 Hermitian matrix is diagonalized, never 16×16.
- The solution is unique, so this *is* the canonical form.
- At $t = 0$: $\mathrm{d}(0) = 0$. The probes first see only the averaged energies $\operatorname{Tr}(\omega_E H_a)$.

Derivation: report, Secs. 4.3–4.5, 4.8, 4.9 and 4.12.

Physical reading: correlated damping and phase loops

Write $\mathrm{d} = S + iA$ (S real symmetric, A real antisymmetric).

- **Damping:** $\kappa_{ab} = \frac{1}{2}\Delta_{ab}^{\mathsf{T}} S \Delta_{ab}$ with $\Delta_{ab} = \mathbf{g}^a - \mathbf{g}^b$; the six κ_{ab} fix S uniquely.
- Independent single-probe dephasing gives $S = \mathrm{diag}(2\kappa^A, 2\kappa^B, 0)$ and $A = 0$. Any other entry signals dephasing *correlated through the bulk*.
- $S_{12} > 0$: $00 \leftrightarrow 11$ decays faster than $01 \leftrightarrow 10$ (common noise). $S_{12} < 0$: anticorrelated noise.
- **Phases:** A is fixed by the three loop sums $\omega_{ab} + \omega_{bc} + \omega_{ca}$. These are phases that no set of four configuration energies can produce.

Derivation: report, Secs. 4.6 and 4.7.

Negative rates, memory and singular times

- Observed damping is bounded by the canonical rates: $\gamma_{\min} \leq \kappa_{ab} \leq \gamma_{\max}$.
- A coherence $|\Gamma_{ab}|$ that *grows* implies a negative canonical rate. The converse fails: $S_{33} < 0$ also gives one.
- CP divisibility $\iff$ $\mathrm{d}(t) \succeq 0$. A negative rate means non-Markovian memory, *not* an unphysical Λ_t, which stays CPTP.
- At a zero of Γ_{ab} the damping rate has a pole and some γ_k diverges, while the phase rate stays bounded.
- $\rho_{AB}(t)$ itself remains smooth throughout.

Derivation: report, Secs. 4.10 and 4.11.

Reference entanglement and probe entanglement

- Unitary iff $\operatorname{rank}\Gamma = 1$, i.e. $\Gamma_{ab} = u_a u_b^*$: a diagonal phase gate, possibly non-local.
- Choi state versus reference: $\mathcal{N}_{AB|R} = \frac{1}{4}\sum_{a<b} |\Gamma_{ab}|$. The channel is entanglement breaking iff every $\Gamma_{ab} = 0$.
- Between the probes:

$$\boxed{\max_{\rho \in \mathrm{Sep}(A:B)} C[\Lambda_t(\rho)] = C[\rho_f(t)].}$$

  The channel can entangle A with B iff its $|{+}{+}\rangle$ output is entangled, and $|{+}{+}\rangle$ is the optimal separable input.

Derivation: report, Secs. 3.7 and 3.9.

Exchange coupling: spin flows, total magnetization survives

$$M_S = S_A^z + S_B^z, \quad M_E = \sum_{j=1}^{N} S_j^z, \qquad [H^{(\mathrm{ex})}, M_S + M_E] = 0.$$

- Each contact moves one unit of S^z between a probe and the bulk.
- Probe populations and coherences can both change.
- There is no four-branch controlled propagator.
- A reduced-channel symmetry also needs a condition on the bulk preparation.

Derivation: report, Sec. 5.1.

Exchange channel: covariance and five blocks

If the bulk preparation has no coherence between magnetization sectors, $[\omega_E, M_E] = 0$, then

$$\boxed{\Lambda_t(e^{-i\theta M_S} X e^{i\theta M_S}) = e^{-i\theta M_S} \Lambda_t(X) e^{i\theta M_S}.}$$

Matrix units $|a\rangle\langle b|$ have coherence order $q = m_a - m_b \in \{-2, \ldots, 2\}$, with probe charges $(1, 0, 0, -1)$. Covariance forbids mixing different q:

q	Size	Content
0	6	Populations and the $01 \leftrightarrow 10$ coherence
± 1	4 each	Single-flip coherences
± 2	1 each	$00 \leftrightarrow 11$ coherence

The Choi matrix has blocks of the same sizes (at most 70 real parameters, not 256). Kraus operators move $0, \pm 1, \pm 2$ units of charge. Derivation: report, Secs. 5.2–5.5.

Fixed total charge gives an X-shaped state

If the *global* state lies in one total-M sector, bulk kets of different charge are orthogonal, and

$$\rho_{AB}(t) = \begin{pmatrix} p_{00} & 0 & 0 & 0 \\ 0 & p_{01} & c & 0 \\ 0 & c^* & p_{10} & 0 \\ 0 & 0 & 0 & p_{11} \end{pmatrix}, \qquad C = 2\max\{0, |c| - \sqrt{p_{00}p_{11}}\}.$$

- Only the $01 \leftrightarrow 10$ coherence survives. It carries all probe–probe entanglement.
- With a fixed-charge bulk, $|{+}{+}\rangle$ occupies *three* total-charge sectors, so its output need not have this form.

Derivation: report, Sec. 5.6.

Quantum-state texture from one measurement setting

Prepare $f_1 = |{+}{+}\rangle\langle{+}{+}|$, the textureless state of the computational basis, and let $\rho_f(t) = \Lambda_t(f_1)$:

$$p_f = \langle{+}{+}|\rho_f|{+}{+}\rangle = \tfrac{1}{4}\sum_{a,b}(\rho_f)_{ab}, \qquad \boxed{\Omega_{T_F} = 1 - p_f, \qquad \Omega_R = -\ln p_f} \quad [2, 3].$$

- p_f is the return probability. Measure both probes along x: $p_f = P^{(x)}_{++}$.
- Equivalently, $4p_f = 1 + \langle\sigma^x_A\rangle + \langle\sigma^x_B\rangle + \langle\sigma^x_A\sigma^x_B\rangle$.
- This holds in both regimes. Texture depends on phases, and is distinct from purity and from entanglement.

Derivation: report, Sec. 2.4.

Ising texture is a channel fidelity

Using $\rho_f = \Gamma/4$:

$$p_f = \frac{1}{4} + \frac{1}{8}\sum_{a<b} \mathrm{Re}\,\Gamma_{ab}, \qquad \boxed{F_e = p_f, \qquad F_{\mathrm{avg}} = \frac{4p_f + 1}{5}} \quad [4].$$

- These are fidelities of the two-probe map relative to the identity. They are not $A \to B$ transfer fidelities.
- The grand-sum response operator is $\mathcal{R}_t = \Gamma^{\mathsf{T}}$.
- $\Omega_{T_F} = 0$ only for the identity channel within the Ising family.
- One real number: it does not fix the six coherences, purity, entanglement or divisibility.

Derivation: report, Secs. 3.10 and 3.11.

Exchange texture: what the same input determines

With a magnetization-invariant bulk, f_1 probes all five coherence orders.

- One output fixes each block's action on *one vector* only. The 6- and 4-dimensional blocks remain undetermined.
- Exception, the one-dimensional sector: $\Lambda_t(|00\rangle\langle 11|) = \xi_{+2}(t)|00\rangle\langle 11|$, with $\boxed{\xi_{+2} = 4[\rho_f]_{00,11}}$.
- $F_e = p_f$ requires the Schur structure; $F_{\mathrm{avg}} = (4F_e + 1)/5$ always holds.

Derivation: report, Sec. 5.7.

Summary

	Ising contacts	Exchange contacts
Map structure	$\Gamma \circ X$, exact	Five coherence-order blocks
Populations	Frozen	Can move
Generator	$\leq$ 3 rates, diagonal L_k, closed form	Block entries from dynamics
Information in ρ_f	Complete channel	Partial block information
Assumptions	Any fixed ω_E, even with transverse fields	$[H, M] = 0$ and $[\omega_E, M_E] = 0$

Symmetry fixes which probe operators can mix. The time dependence of the coefficients requires the many-body bulk dynamics.

Conventions

- Probe entropy $S(\rho) = -\operatorname{Tr}\rho\ln\rho$ in nats. For a globally pure state, $S(\rho_{AB})$ measures entanglement across $AB|E$.
- Concurrence and negativity concern $A|B$. For two qubits, $C > 0 \iff \mathcal{N} > 0$ [5].
- The normalized Choi state has trace one. Its reference–output entanglement is distinct from entanglement between the probes.
- Notation: Γ_{ab} coherence factors, η_{ab} their logarithmic derivatives, γ_k canonical rates, $\xi_{\pm 2}$ exchange multipliers.

References I

[1] Michael J. W. Hall, James D. Cresser, Li Li, and Erika Andersson. Canonical form of master equations and characterization of non-Markovianity. *Phys. Rev. A*, 89(4):042120, 2014. doi:10.1103/PhysRevA.89.042120.

[2] Fernando Parisio. Quantum-State Texture and Gate Identification. *Phys. Rev. Lett.*, 133:260801, 2024. doi:10.1103/PhysRevLett.133.260801. Defines the textureless state $|f_1\rangle$, the grand sum and the rugosity $R = -\ln\langle f_1|\rho|f_1\rangle$.

[3] Carlos H. S. Vieira, Xinfang Nie, Dawei Lu, and Fernando Parisio. Quantum-State Texture Dynamics: Theory and Experiment. arXiv:2609.05248, 2026. URL https://arxiv.org/abs/2609.05248. Free maps obey $\Lambda(f_1) = f_1$; texture preservation iff f_1 is a fixed point of the dual map (Prop. 2).

[4] Michael A. Nielsen. A simple formula for the average gate fidelity of a quantum dynamical operation. *Phys. Lett. A*, 303(4):249–252, 2002. doi:10.1016/S0375-9601(02)01272-0.

[5] William K. Wootters. Entanglement of Formation of an Arbitrary State of Two Qubits. *Phys. Rev. Lett.*, 80(10):2245–2248, 1998. doi:10.1103/PhysRevLett.80.2245.